

Enthusiast Course

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2022

Test Type : **PRACTICE TEST**

Test Pattern : **NEET (UG)**

TEST DATE :

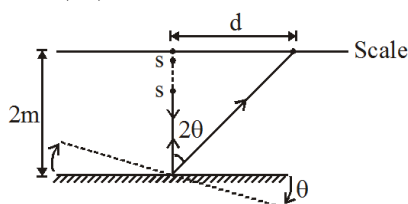
ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	4	4	3	1	3	4	4	4	1	4	4	3	2	4	2	2	4	1	3	4	3	3	2	4	2	1	4	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	3	1	3	2	2	1	4	4	4	2	1	1	1	2	2															

HINT - SHEET



1. **Ans (2)**



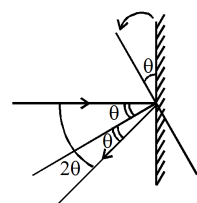
$$2\theta = \frac{d}{2}$$

$$2 \times 1.8^\circ = \frac{d}{2}$$

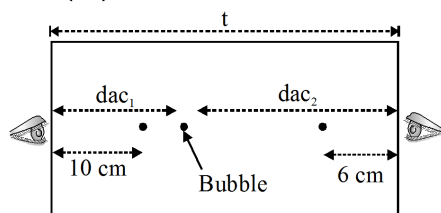
$$d = 4 \times 1.8^\circ$$

$$d = 4 \times 1.8 \times \frac{\pi}{180} = 0.1256 \text{ m}$$

2. **Ans (4)**



4. **Ans (3)**



$$t = dac_1 + dac_2$$

$$= \mu \times dap_1 + \mu \times dap_2$$

$$= \frac{3}{2}(10 + 6) = \frac{3}{2} \times 16 = 24 \text{ cm}$$

5. **Ans (1)**

$$P = \frac{100}{v} - \frac{100}{u}$$

$$P = \frac{100}{-200} - \frac{100}{-\infty}$$

$$P = -0.5 \text{ D concave lens}$$

6. **Ans (3)**

$$\text{Final image } v_c \rightarrow \infty$$

$$L = v_0 + f_e$$

$$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\frac{1}{v_0} = \frac{1}{-400} - \frac{1}{+50}$$

$$\frac{1}{v_0} = \frac{1}{50} - \frac{1}{400} = \frac{8-1}{400}$$

$$v_0 = \frac{400}{7} \text{ cm}$$

$$L = \frac{400}{7} + 5$$

$$L = \frac{435}{7} \text{ cm}$$

$$L = 62.14 \text{ cm}$$

7. **Ans (4)**

$$\delta = 2i - A$$

$$= 2 \times 30^\circ - 45^\circ$$

$$= 60^\circ - 45^\circ = 15^\circ$$

9. **Ans (4)**

$$\mu = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \cos \text{ec} \frac{A}{2} = \frac{1}{\sin \frac{A}{2}}$$

$$\sin\left(\frac{\delta_m + A}{2}\right) = 1$$

$$\frac{\delta_m + A}{2} = 90^\circ$$

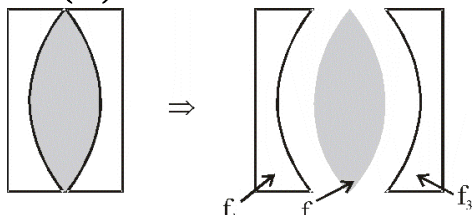
$$\delta_m = 180^\circ - A$$

10. **Ans (1)**

$$\mu = \frac{1}{\sin i_C} = \frac{1}{\sin 30^\circ} = 2$$

no colour will suffer TIR

11. **Ans (4)**



$$f_1 = f_3 = -\frac{R}{(\mu - 1)} = \frac{-10}{\frac{3}{2} - 1} = -20 \text{ cm}$$

$$f_2 = \frac{R}{2(\mu - 1)} = +\frac{10}{2\left(\frac{4}{3} - 1\right)} = +15 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

$$\frac{1}{f} = \frac{-1}{20} + \frac{1}{15} - \frac{1}{20} = -\frac{1}{10} + \frac{1}{15}$$

$$\frac{1}{f} = \frac{-3 + 2}{30}$$

$$f = -30 \text{ cm}$$

12. **Ans (4)**

$$P = P_1 + P_2 = 20 + 10 = 30 \text{ D}$$

$$P = \frac{100}{v} - \frac{100}{u}$$

$$30 = \frac{100}{v} - \frac{100}{-\infty}$$

$$v = \frac{100}{30} = 3.33 \text{ cm}$$

13. **Ans (3)**

Minimum deviation, $A = 2r$

$$r = \frac{A}{2} = \frac{30^\circ}{2} = 15^\circ$$

14. **Ans (2)**

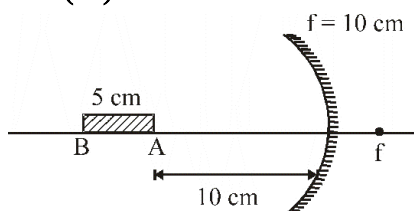


Image of one end of rod will form at infinity
so length of image = ∞

15. **Ans (4)**

$$M.P. = -\frac{f_0}{f_e} = -7$$

$$f_0 = 7f_e$$

$$f_0 = 7f_e$$

$$L = f_0 + f_e = 40$$

$$7f_e + f_e = 40$$

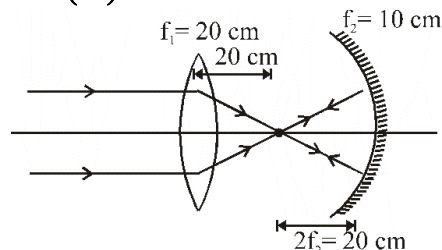
$$8f_e = 40$$

$$f_e = 5 \text{ cm}$$

$$f_0 = 7f_e = 7 \times 5 = 35 \text{ cm}$$

Tele: @chalnaayaaaar_bot

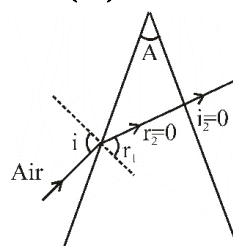
16. **Ans (2)**



$$d = -2f_2 + f_1 = -2(-10) + 20 = 20 + 20$$

$$d = 40 \text{ cm}$$

17. **Ans (2)**



$$A = r_1 + r_2$$

$$4^\circ = r_1 + 0$$

$$r_1 = 4^\circ$$

$$1 \times \sin i = \mu \times \sin r_1$$

$$\sin i = 1.5 \times \sin 4^\circ$$

$$i = 6^\circ$$

18. **Ans (4)**

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{20} - \frac{1}{12} = \frac{1}{f}$$

$$\frac{1}{20} - \frac{1}{12} = \frac{1}{f} = \frac{12 - 20}{240}$$

$$f = -\frac{240}{8} = -30 \text{ cm}$$

19. **Ans (1)**

$$(\mu - 1)A + (\mu' - 1)A' = 0$$

$$(1.5 - 1) \times 8^\circ + (1.64 - 1)A' = 0$$

$$A' = \frac{-0.5 \times 8^\circ}{0.64} = -6.25^\circ$$

20. **Ans (3)**

$$f = \frac{R}{2(\mu - 1)} = \frac{40}{2(1.5 - 1)} = \frac{20}{0.5} = 40 \text{ cm}$$

$$\frac{h_1}{h_0} = \frac{f}{f + 60}$$

$$\frac{h_1}{4} = \frac{40}{40 - 60} = \frac{40}{-20} = -2$$

$$h_1 = -8 \text{ cm}$$

21. **Ans (4)**

$$f_1 = \frac{R}{(\mu - 1)} = \frac{10}{(1.47 - 1)}$$

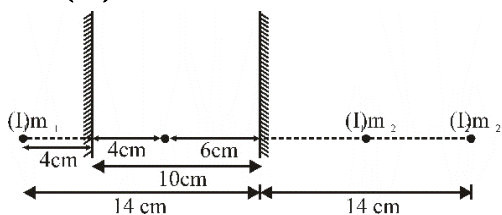
$$f_2 = -\frac{R}{(\mu - 1)} = -\frac{10}{(1.74 - 1)}$$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{0.47}{10} - \frac{0.74}{10}$$

$$\Rightarrow \frac{1}{f} = -\frac{0.27}{10}$$

$$f = \frac{10}{-0.27} = -37.03 \text{ cm}$$

22. **Ans (3)**



Distance between first image formed by M_1 and second image formed by $M_2 = 28 \text{ cm}$

23. **Ans (3)**

$f = R/2$ by formula

$$u = f \left(1 - \frac{1}{m} \right) = 17.5 \left(1 - \frac{1}{2.5} \right) = 10.5 \text{ cm}$$

25. **Ans (4)**

$$r = \frac{h}{\sqrt{\mu^2 - 1}} = \frac{12}{\sqrt{\left(\frac{4}{3}\right)^2 - 1}} = \frac{12 \times 3}{\sqrt{7}}$$

26. **Ans (2)**

$$2(\mu - 1) \frac{2}{R} + \frac{2}{R}$$

$$2 \times \frac{1}{2} \times \frac{2}{R} + \frac{2}{R}$$

$$\Rightarrow 5$$

27. **Ans (1)**

Distance between lenses $L = V_0 + u_e$

for objective lens

$$\frac{1}{V_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\Rightarrow \frac{1}{V_0} - \frac{1}{(-3.75)} = \frac{1}{2.5}$$

$$\Rightarrow V_0 = 7.5 \text{ cm}$$

for eye lens

$$\frac{1}{V_e} - \frac{1}{u_e} = \frac{1}{f_e} \Rightarrow \frac{1}{(-25)} - \frac{1}{u_e} = \frac{1}{5}$$

$$\Rightarrow u_e = 4.166$$

$$L = 11.67 \text{ cm}$$

28. **Ans (4)**

Focused distance of camera

= Distance of camera from mirror

+ Distance of image from mirror

$$= 4.5 \text{ m} + 3 \text{ m}$$

$$= 7.5 \text{ m}$$

29. **Ans (1)**

For concave mirror

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{(-12)} + \frac{1}{v} = \frac{1}{(-9)}$$

$$\Rightarrow v = -36$$

magnification

$$m = -\frac{v}{u} \Rightarrow -\frac{(-36)}{(-12)} \Rightarrow m = -3$$

$$m = \frac{h_i}{h_o} = -3 \Rightarrow h_i = -3(h_o)$$

$$= -9 \text{ cm}$$

$$= \text{Real, } 9 \text{ cm}$$

30. **Ans (3)**

$$d_{\text{app}} = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} \Rightarrow d_{\text{app}} = \frac{30}{1.5} + \frac{30}{2}$$

$$d_{\text{app}} = 35 \text{ cm}$$

31. **Ans (3)**

$$\lambda \propto v \propto \frac{1}{\mu}$$

$$\frac{\lambda_{\text{water}}}{\lambda_{\text{air}}} = \frac{\mu_{\text{air}}}{\mu_{\text{water}}} \Rightarrow \lambda_{\text{water}} = \frac{1}{(4/3)} \times 4200 \text{ \AA}$$

$$= 3150 \text{ \AA}$$

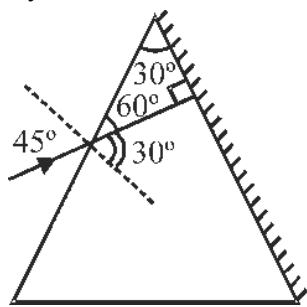
32. **Ans (1)**

$$t = \frac{d}{v} \Rightarrow t = \frac{10 \text{ mm}}{\left(\frac{c}{\mu}\right)}$$

$$t = \frac{10 \times 10^{-3} \times 3}{2 \times 3 \times 10^8} \Rightarrow t = 0.05 \text{ n sec}$$

33. Ans (3)

By shell's law



$$1 \times \sin 45^\circ = \mu \sin 30^\circ$$

$$\mu = \sqrt{2}$$

35. Ans (2)

$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

$$\Rightarrow \frac{1}{f_{eq}} = \frac{(1.5 - 1)}{12} - \frac{2(\mu - 1)}{12} + \frac{(1.5 - 1)}{12}$$

$$\frac{1}{60} = \frac{1}{12} - \frac{(\mu - 1)}{6} \Rightarrow \mu = 1.4$$

37. Ans (4)

$$M = 1 + \frac{D}{f}$$

$$\Rightarrow M = 1 + \frac{25}{5}$$

$$\Rightarrow M = 6$$

38. Ans (4)

Angle of deviation for prism

$$\delta = i + e - A$$

$$\Rightarrow 42 = 50 + e - 60$$

$$\Rightarrow e = 52^\circ$$

$$\text{Angle from surface} = 90 - 52 = 38^\circ$$

39. Ans (4)

Resolving power of telescope

$$R.P. = \frac{D}{1.22\lambda}$$

$$R.P. = \frac{0.61}{1.22 \times 5000 \times 10^{-10}}$$

$$\Rightarrow R.P. = 10^6$$

40. Ans (2)

Here angle of emergence

= angle of incidence

So, angle of divergence of emergent beam

= angle of divergence of incident beam

$$= \alpha$$

41. Ans (1)

$$\text{For lens } \frac{1}{V} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{V} - \frac{1}{-30} = \frac{1}{20}$$

$$\Rightarrow V = 60 \text{ cm}$$

Final image coincide with object so light ray must retrace its path.

$$\text{Radius of curvature of mirror} = V - 20$$

$$= 60 - 20$$

$$= 40$$

$$\text{Focal length of mirror} = \frac{R}{2}$$

$$\Rightarrow 20 \text{ cm}$$

42. Ans (1)

$$\text{Magnification } M = \frac{V_0}{u_0} \left(1 + \frac{D}{f_e} \right)$$

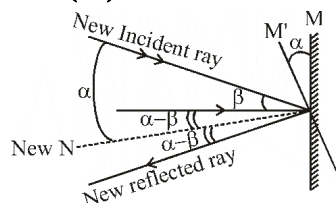
$$\text{for objective } \frac{1}{V_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\Rightarrow \frac{1}{V_0} - \frac{1}{(-5)} = \frac{1}{4}$$

$$\Rightarrow V_0 = 20 \text{ cm}$$

$$M = \frac{20}{5} \left(1 + \frac{20}{10} \right) \Rightarrow M = 12$$

43. Ans (1)



angle between original incident ray

and new reflected ray = $\alpha + \alpha - \beta = 2\alpha - \beta$

44. Ans (2)

$$(\mu - 1)A + (\mu' - 1)A' = 0$$

$$(1.52 - 1) \times 10 + (1.72 - 1)A' = 0$$

$$A' = -\frac{0.52 \times 10}{0.72} = -7.22$$

45. Ans (2)

$$f_1 = f_3 = \frac{R}{(\mu - 1)} = \frac{10}{\left(\frac{3}{2} - 1\right)} = \frac{10}{\frac{1}{2}} = 20 \text{ cm}$$

$$f_2 = -\frac{R}{2(\mu - 1)} = -\frac{10}{2\left(\frac{4}{3} - 1\right)} = -\frac{5}{\frac{1}{3}} = -15 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} = \frac{1}{20} - \frac{1}{15} + \frac{1}{20} = \frac{1}{10} - \frac{1}{15}$$

$$= \frac{3 - 2}{30}$$

$$F = 30 \text{ cm}$$